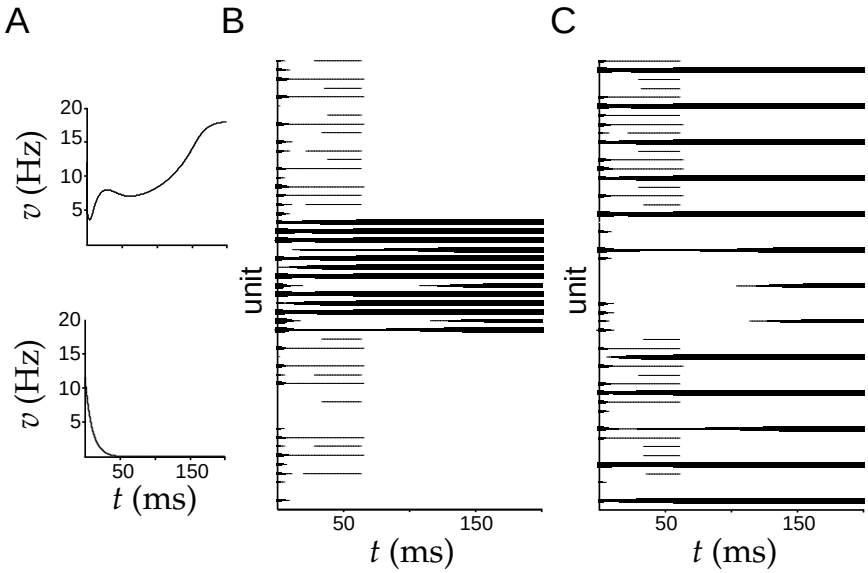

Figure 7.16 Associative recall of memory patterns in a network model. Panel A shows two representative units, and panels B and C show the firing rates of all 50 units plotted against time. The thickness of the horizontal lines in these plots is proportional to the firing rate of the corresponding neuron. (A) Firing rates of representative neurons. The upper panel shows the firing rate of one of the neurons corresponding to a nonzero component of the recalled memory pattern. The firing rate achieves a nonzero (and nonsaturated) steady-state value. The lower panel shows the firing rate of a neuron corresponding to a zero component of the recalled memory pattern. This goes to 0. (B) Recall of one of the stored memory patterns. The stored pattern had nonzero values for units 18 through 31. The initial state of the network was random, but with a bias toward this particular pattern. The final state is similar to the memory pattern. (C) Recall of another of the stored memory patterns. The stored pattern had nonzero values for every fourth unit. The initial state of the network was again random, but biased toward this pattern. The final state is similar to the memory pattern. This model uses the matrix of equation 7.48 with $\alpha = 0.25$ and $\lambda = 1.25$, and the activation function $F(I) = 150\,\text{Hz}[\tanh((I + 20\,\text{Hz})/(150\,\text{Hz}))]_+$.

7.5 Excitatory-Inhibitory Networks

In this section, we discuss models in which excitatory and inhibitory neurons are described separately by equations 7.12 and 7.13. These models exhibit richer dynamics than the single population models with symmetric coupling matrices we have analyzed up to this point. In models with excitatory and inhibitory subpopulations, the full synaptic weight matrix is not symmetric, and network oscillations can arise. We begin by analyzing a model of homogeneous coupled excitatory and inhibitory populations. We introduce methods for determining whether this model exhibits constant or oscillatory activity. We then present two network models in which oscillations appear. The first is a model of the olfactory bulb, and the second displays selective amplification in an oscillatory mode.

Homogeneous Excitatory and Inhibitory Populations

As an illustration of the dynamics of excitatory-inhibitory network models, we analyze a simple model in which all of the excitatory neurons are described by a single firing rate, v_E, and all of the inhibitory neurons are described by a second rate, v_I. Although we think of this example as a model of interacting neuronal populations, it is constructed as if it consists of just two neurons. Equations 7.12 and 7.13, with threshold linear response functions, are used to describe the two firing rates, so that

$$\tau_E \frac{dv_E}{dt} = -v_E + [M_{EE}v_E + M_{EI}v_I - \gamma_E]_+ \tag{7.50}$$

and

$$\tau_I \frac{dv_I}{dt} = -v_I + [M_{II}v_I + M_{IE}v_E - \gamma_I]_+ \ . \tag{7.51}$$

The synaptic weights M_{EE}, M_{IE}, M_{EI}, and M_{II} are numbers rather than matrices in this model. In the example we consider, we set $M_{EE} = 1.25$, $M_{IE} = 1$, $M_{II} = 0$, $M_{EI} = -1$, $\gamma_E = -10$ Hz, $\gamma_I = 10$ Hz, $\tau_E = 10$ ms, and we vary the value of τ_I. The negative value of γ_E means that this parameter serves as a source of constant background activity rather than as a threshold.

Phase-Plane Methods and Stability Analysis

The model of interacting excitatory and inhibitory populations given by equations 7.50 and 7.51 provides an opportunity for us to illustrate some of the techniques used to study the dynamics of nonlinear systems. This model exhibits both fixed-point (constant v_E and v_I) and oscillatory activity, depending on the values of its parameters. Stability analysis can be used to determine the parameter values where transitions between these two types of activity take place.

The firing rates $v_E(t)$ and $v_I(t)$ arising from equations 7.50 and 7.51 can be displayed by plotting them as functions of time, as in figures 7.18A and 7.19A. Another useful way of depicting these results, illustrated in figures 7.18B and 7.19B, is to plot pairs of points $(v_E(t), v_I(t))$ for a range of t values. As the firing rates change, these points trace out a curve or trajectory in the v_E-v_I plane, which is called the phase plane of the model. Phase-plane plots can be used to give a geometric picture of the dynamics of a model.

phase plane

Values of v_E and v_I for which the right side of either equation 7.50 or equation 7.51 vanishes are of particular interest in phase-plane analysis. Sets of such values form two curves in the phase plane known as nullclines. The nullclines for equations 7.50 and 7.51 are the straight lines drawn in figure 7.17A. The nullclines are important because they divide the phase plane into regions with opposite flow patterns. This is because dv_E/dt and

nullcline

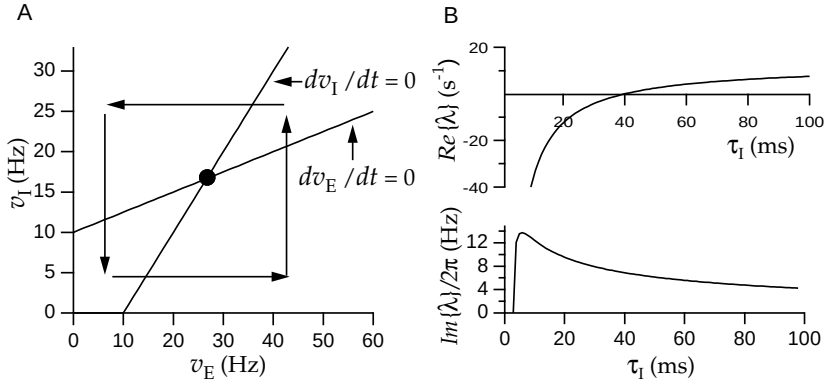

Figure 7.17 (A) Nullclines, flow directions, and fixed point for the firing-rate model of interacting excitatory and inhibitory neurons. The two straight lines are the nullclines along which $dv_E/dt = 0$ or $dv_I/dt = 0$. The filled circle is the fixed point of the model. The horizontal and vertical arrows indicate the directions that v_E (horizontal arrows) and v_I (vertical arrows) flow in different regions of the phase plane relative to the nullclines. (B) Real (upper panel) and imaginary (lower panel) parts of the eigenvalue determining the stability of the fixed point. To the left of the point where the imaginary part of the eigenvalue goes to 0, both eigenvalues are real. The imaginary part has been divided by 2π to give the frequency of oscillations near the fixed point.

dv_I/dt are positive on one side of their nullclines and negative on the other, as the reader can verify from equations 7.50 and 7.51 Above the nullcline along which $dv_E/dt = 0$, $dv_E/dt < 0$, and below it $dv_E/dt > 0$. Similarly, $dv_I/dt > 0$ to the right of the nullcline where $dv_I/dt = 0$, and $dv_I/dt < 0$ to the left of it. This determines the direction of flow in the phase plane, as denoted by the horizontal and vertical arrows in figure 7.17A. Furthermore, the rate of flow typically slows if the phase-plane trajectory approaches a nullcline.

At a fixed point of a dynamic system, the dynamic variables remain at constant values. In the model being considered, a fixed point occurs when the firing rates v_E and v_I take values that make $dv_E/dt = dv_I/dt = 0$. Because a fixed point requires both derivatives to vanish, it can occur only at an intersection of nullclines. The model we are considering has a single fixed point (at $v_E = 26.67$, $v_I = 16.67$) denoted by the filled circle in figure 7.17A. A fixed point provides a potential static configuration for the system, but it is critically important whether the fixed point is stable or unstable. If a fixed point is stable, initial values of v_E and v_I near the fixed point will be drawn toward it over time. If the fixed point is unstable, nearby configurations are pushed away from the fixed point, and the system will remain at the fixed point indefinitely only if the rates are set initially to the fixed-point values with infinite precision.

Linear stability analysis can be used to determine whether a fixed point is stable or unstable. To do this we take derivatives of the expressions for dv_E/dt and dv_I/dt obtained by dividing the right sides of equations 7.50

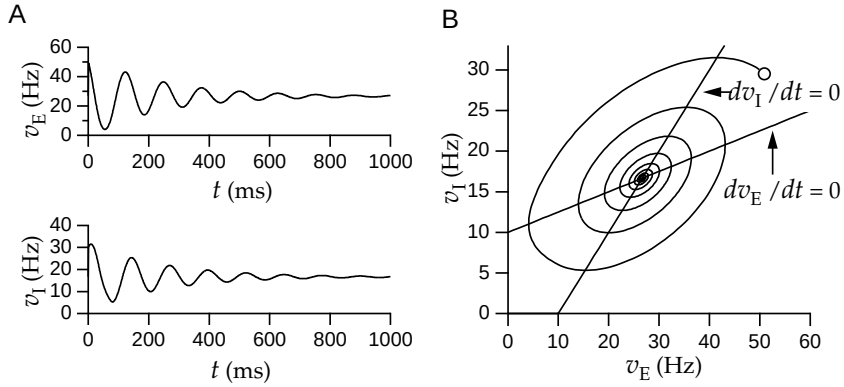

Figure 7.18 Activity of the excitatory-inhibitory firing-rate model when the fixed point is stable. (A) The excitatory and inhibitory firing rates settle to the fixed point over time. (B) The phase-plane trajectory is a counterclockwise spiral collapsing to the fixed point. The open circle marks the initial values $v_E(0)$ and $v_I(0)$. For this example, $\tau_I = 30$ ms.

and 7.51 by τ_E and τ_I respectively. We then evaluate these derivatives at the values of v_E and v_I that correspond to the fixed point. The four combinations of derivatives computed in this way can be arranged into a matrix

stability matrix

$$\begin{pmatrix} (M_{EE}-1)/\tau_E & M_{EI}/\tau_E \\ M_{IE}/\tau_I & (M_{II}-1)/\tau_I \end{pmatrix}. \tag{7.52}$$

As discussed in the Mathematical Appendix, the stability of the fixed point is determined by the real parts of the eigenvalues of this matrix. The eigenvalues are given by

$$\lambda = \frac{1}{2}\left(\frac{M_{EE}-1}{\tau_E} + \frac{M_{II}-1}{\tau_I} \pm \sqrt{\left(\frac{M_{EE}-1}{\tau_E} - \frac{M_{II}-1}{\tau_I}\right)^2 + \frac{4M_{EI}M_{IE}}{\tau_E\tau_I}}\right). \tag{7.53}$$

If the real parts of both eigenvalues are less than 0, the fixed point is stable, whereas if either is greater than 0, the fixed point is unstable. If the factor under the radical sign in equation 7.53 is positive, both eigenvalues are real, and the behavior near the fixed point is exponential. This means that there is exponential movement toward the fixed point if both eigenvalues are negative, or away from the fixed point if either eigenvalue is positive. We focus on the case when the factor under the radical sign is negative, so that the square root is imaginary and the eigenvalues form a complex conjugate pair. In this case, the behavior near the fixed point is oscillatory and the trajectory either spirals into the fixed point, if the real part of the eigenvalues is negative, or out from the fixed point if the real part of the eigenvalues is positive. The imaginary part of the eigenvalue determines the frequency of oscillations near the fixed point. The real and imaginary parts of one of these eigenvalues are plotted as a function of τ_I

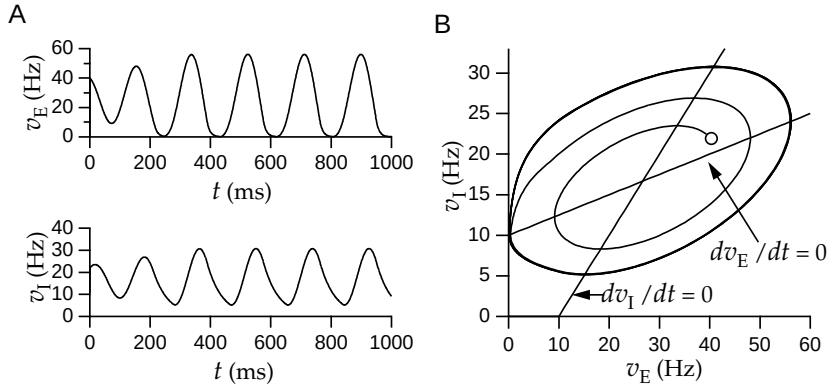

Figure 7.19 Activity of the excitatory-inhibitory firing-rate model when the fixed point is unstable. (A) The excitatory and inhibitory firing rates settle into periodic oscillations. (B) The phase-plane trajectory is a counterclockwise spiral that joins the limit cycle, which is the closed orbit. The open circle marks the initial values $v_E(0)$ and $v_I(0)$. For this example, $\tau_I = 50$ ms.

in figure 7.17B. This figure indicates that the fixed point is stable if $\tau_I < 40$ ms and unstable for larger values of τ_I.

Figures 7.18 and 7.19 show examples in which the fixed point is stable and unstable, respectively. In figure 7.18A, the oscillations in v_E and v_I are damped, and the firing rates settle down to the stable fixed point. The corresponding phase-plane trajectory is a collapsing spiral (figure 7.18B). In figure 7.19A the oscillations grow, and in figure 7.19B the trajectory is a spiral that expands outward until the system enters a limit cycle. A limit cycle is a closed orbit in the phase plane indicating periodic behavior. The fixed point is unstable in this case, but the limit cycle is stable. Without rectification, the phase-plane trajectory would spiral out from the unstable fixed point indefinitely. The rectification nonlinearity prevents the spiral trajectory from expanding past 0 and thereby stabilizes the limit cycle.

limit cycle

There are a number of ways that a nonlinear system can make a transition from a stable fixed point to a limit cycle. Such transitions are called bifurcations. The transition seen between figures 7.18 and 7.19 is a Hopf bifurcation. In this case, a fixed point becomes unstable as a parameter is changed (in this case τ_I) when the real part of a complex eigenvalue changes sign. In a Hopf bifurcation, the limit cycle emerges at a finite frequency, which is similar to the behavior of a type II neuron when it starts firing action potentials, as discussed in chapter 6. Other types of bifurcations produce type I behavior with oscillations emerging at 0 frequency (chapter 6). One example of this is a saddle-node bifurcation, which occurs when parameters are changed such that two fixed points, one stable and one unstable, meet at the same point in the phase plane.

Hopf bifurcation

saddle-node bifurcation

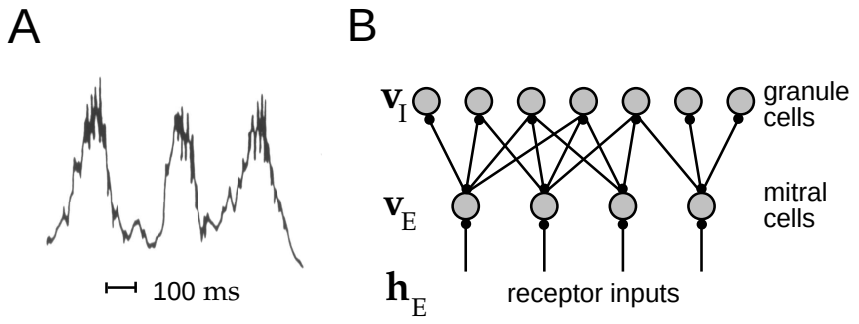

Figure 7.20 (A) Extracellular field potential recorded in the olfactory bulb during respiratory waves representing three successive sniffs. (B) Schematic diagram of the olfactory bulb model. (A adapted from Freeman and Schneider, 1982; B adapted from Li, 1995.)

The Olfactory Bulb

The olfactory bulb, and analogous olfactory areas in insects, provide examples of sensory processing involving oscillatory activity. The olfactory bulb represents the first stage of processing beyond the olfactory receptors in the vertebrate olfactory system. Olfactory receptor neurons respond to odor molecules and send their axons to the olfactory bulb. These axons terminate in glomeruli where they synapse onto mitral and tufted cells, as well as local interneurons. The mitral and tufted cells provide the output of the olfactory bulb by sending projections to the primary olfactory cortex. They also synapse onto the larger population of inhibitory granule cells. The granule cells in turn inhibit the mitral and tufted cells.

mitral cells
tufted cells
granule cells

The activity in the olfactory bulb of many vertebrates is strongly influenced by a sniff cycle in which a few quick sniffs bring odors past the olfactory receptors. Figure 7.20A shows an extracellular potential recorded during three successive sniffs. The three large oscillations in the figure are due to the sniffs. The oscillations we discuss in this section are the smaller, higher-frequency oscillations seen around the peak of each sniff cycle. These arise from oscillatory neural activity. Individual mitral cells have quite low firing rates, and do not fire on each cycle of the oscillations. The oscillations are phase-locked across the bulb, in that different neurons fire at fixed phase lags from each other, but different odors induce oscillations of different amplitudes and phases.

Li and Hopfield (1989) modeled the mitral and granule cells of the olfactory bulb as a nonlinear input-driven network oscillator. Figure 7.20B shows the architecture of the model, which uses equations 7.12 and 7.13 with $\mathbf{M}_{EE} = \mathbf{M}_{II} = 0$. The absence of these couplings in the model is in accord with the anatomy of the bulb. The rates $\mathbf{v}_E$ and $\mathbf{v}_I$ refer to the mitral and granule cells, respectively. Figure 7.21A shows the activation functions of the model. The time constants for the two populations of cells are the same, $\tau_E = \tau_I = 6.7$ ms. $\mathbf{h}_E$ is the input from the receptors to the mitral

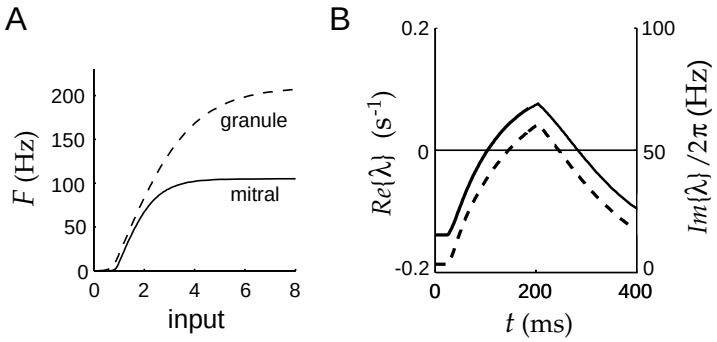

Figure 7.21 Activation functions and eigenvalues for the olfactory bulb model. (A) The activation functions F_E (solid curve) for the mitral cells, and F_I (dashed curve) for the granule cells. (B) The real (solid line, left axis) and imaginary (dashed line, right axis) parts of the eigenvalue that determines whether the network model exhibits fixed-point or oscillatory behavior. These are plotted as a function of time during a sniff cycle. When the real part of the eigenvalue becomes greater than 0, it determines the growth rate away from the fixed point, and the imaginary part divided by 2π determines the initial frequency of the resulting oscillations. (Adapted from Li, 1995.)

cells, and $\mathbf{h}_I$ is a constant representing top-down input that exists from the olfactory cortex to the granule cells.

The field potential in figure 7.20A shows oscillations during each sniff, but not between sniffs. For the model to match this pattern of activity, the input from the olfactory receptors, $\mathbf{h}_E$, must induce a transition between fixed-point and oscillatory activity. Before a sniff, the network must have a stable fixed point with low activities. As $\mathbf{h}_E$ increases during a sniff, this steady-state configuration must become unstable, leading to oscillatory activity. The analysis of the stability of the fixed point and the onset of oscillations is closely related to our previous stability analysis of the model of homogeneous populations of coupled excitatory and inhibitory neurons. It is based on properties of the eigenvalues of the linear stability matrix (see the Mathematical Appendix). In this case, the stability matrix includes contributions from the derivatives of the activation functions evaluated at the fixed point. For the fixed point to become unstable, the real part of at least one of the eigenvalues that arise in this analysis must become larger than 0. To ensure oscillations, at least one of these destabilizing eigenvalues should have a nonzero imaginary part. These requirements impose constraints on the connections between the mitral and granule cells and on the inputs.

Figure 7.21B shows the real and imaginary parts of the relevant eigenvalue, labeled λ, during one sniff cycle. About 100 ms into the cycle the real part gets bigger than 1. Reading off the imaginary part at this point, we see that this sets off roughly 40 Hz oscillations in the network. These oscillations stop about 300 ms into the sniff cycle when the real part of λ drops below 0. The input $\mathbf{h}_E$ from the receptors plays two critical roles in this process. First, it makes the real part of the eigenvalue greater than 0 by

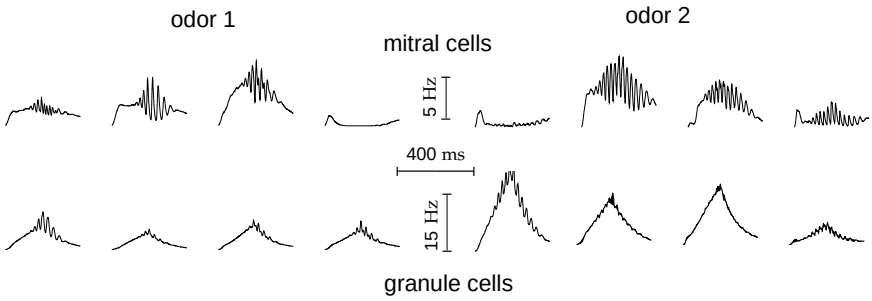

Figure 7.22 Activities of four of ten mitral (upper) and granule (lower) cells during a single sniff cycle for two different odors. (Adapted from Li and Hopfield, 1989.)

modifying where the fixed point lies on the activation function curves in figure 7.21A. Second, it affects which particular neurons are destabilized, and thus which begin to oscillate. The ultimate pattern of oscillatory activity is determined both by the input $\mathbf{h}_{\mathrm{E}}$ and by the recurrent couplings of the network.

Figure 7.22 shows the behavior of the network during a single sniff cycle in the presence of two different odors, represented by two different values of $\mathbf{h}_{\mathrm{E}}$. The top rows show the activity of four mitral cells, and the bottom rows of four granule cells. The amplitudes and phases of the oscillations seen in these traces, along with the identities of the mitral cells taking part in them, provide a signature of the identity of the odor that was presented.

Oscillatory Amplification

As a final example of network oscillations, we return to amplification of input signals by a recurrently connected network. Two factors control the amount of selective amplification that is viable in networks such as that shown in figure 7.9. The most important constraint on the recurrent weights is that the network must be stable, so the activity does not increase without bound. Another possible constraint is suggested by figure 7.14D, where the output shows a tuned response even though the input to the network is constant as a function of θ. Tuned output in the absence of tuned input can serve as a memory mechanism, but it will produce persistent perceptions if it occurs in a primary sensory area, for example. Avoiding this in the network limits the recurrent weights and the amount of amplification that can be supported.

Li and Dayan (1999) showed that this restriction can be significantly eased using the richer dynamics of networks of coupled inhibitory and excitatory neurons. Figure 7.23 shows an example with continuous neuron labeling based on a continuous version of equations 7.12 and 7.13. The input is $h_{\mathrm{E}}(\theta) = 8 + 5\cos(2\theta)$ in the modulated case (figure 7.23B) or $h_{\mathrm{E}}(\theta) = 8$ in the unmodulated case (figure 7.23C). Noise with standard deviation 0.4 corrupts this input. The input to the network is constant in time.

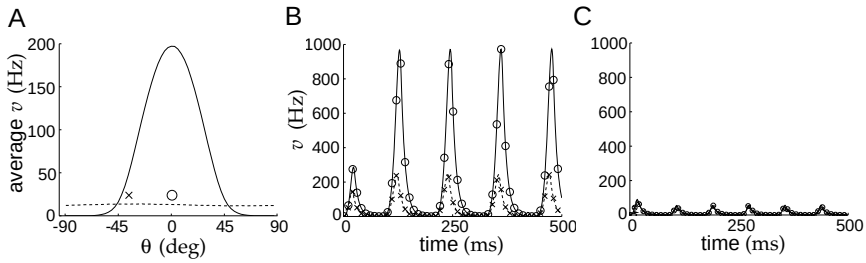

Figure 7.23 Selective amplification in an excitatory-inhibitory network. (A) Time-averaged response of the network to a tuned input with $\Theta = 0^\circ$ (solid curve) and to an untuned input (dashed curve). Symbols "o" and "x" mark the 0° and -37° points seen in B and C. (B) Activities over time of neurons with preferred angles of $\theta = 0^\circ$ (solid curve) and $\theta = -37^\circ$ (dashed curve) in response to a modulated input with $\Theta = 0^\circ$. (C) Activities of the same units shown in B to a constant input. The lines lie on top of each other, showing that the two units respond identically. The parameters are $\tau_E = \tau_I = 10$ ms, $h_I = 0$, $M_{EI} = -\delta(\theta - \theta')/\rho_\theta$, $M_{EE} = (1/\pi\rho_\theta)[5.9 + 7.8\cos(2(\theta - \theta'))]_+$, $M_{IE} = 13.3/\pi\rho_\theta$, and $M_{II} = 0$. (After Li and Dayan, 1999.)

The network oscillates in response to either constant or tuned input. Figure 7.23A shows the time average of the oscillating activities of the neurons in the network as a function of their preferred angles for noisy tuned (solid curve) and untuned (dashed curve) inputs. Neurons respond to the tuned input in a highly tuned and amplified manner. Despite the high degree of amplification, the average response of the neurons to untuned input is almost independent of θ. Figures 7.23B and 7.23C show the activities of individual neurons with $\theta = 0^\circ$ ("o") and $\theta = -37^\circ$ ("x") over time for the tuned and untuned inputs respectively. The network does not produce persistent perception, because the output to an untuned input is itself untuned. In contrast, a nonoscillatory version of this network, with $\tau_I = 0$, exhibits tuned sustained activity in response to an untuned input for recurrent weights this strong. The oscillatory network can thus operate in a regime of high selective amplification without generating spurious tuned activity.

7.6 Stochastic Networks

Up to this point, we have considered models in which the output of a cell is a deterministic function of its input. In this section, we introduce a network model called the Boltzmann machine, for which the input-output relationship is stochastic. Boltzmann machines are interesting from the perspective of learning, and also because they offer an alternative interpretation of the dynamics of network models.

Boltzmann machine

In the simplest form of Boltzmann machine, the neurons are treated as binary, so $v_a(t) = 1$ if unit a is active at time t (e.g., it fires a spike between times t and $t + \Delta t$ for some small value of Δt), and $v_a(t) = 0$ if it is inactive.